

IOT BASED SMART FALL DETECTION FOR ELDERLY PEOPLE

Electronics & Telecommunication Engineering

 Domain :-IoT · Healthcare · Embedded Systems

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Why Elderly Fall Detection Matters

Falls are the leading cause of injury-related deaths among adults aged 65+. With global aging populations rising rapidly, IoT-enabled wearables offer a life-saving bridge between incident and emergency response.

36M

Falls Per Year
Reported annually in
the US among elderly
adults

32K

Annual
Deaths
Fall-related fatalities in
seniors each year

5s

Critical Window
Response time that can
determine survival
outcomes

Problem Statement

⚠ High Fall Risk

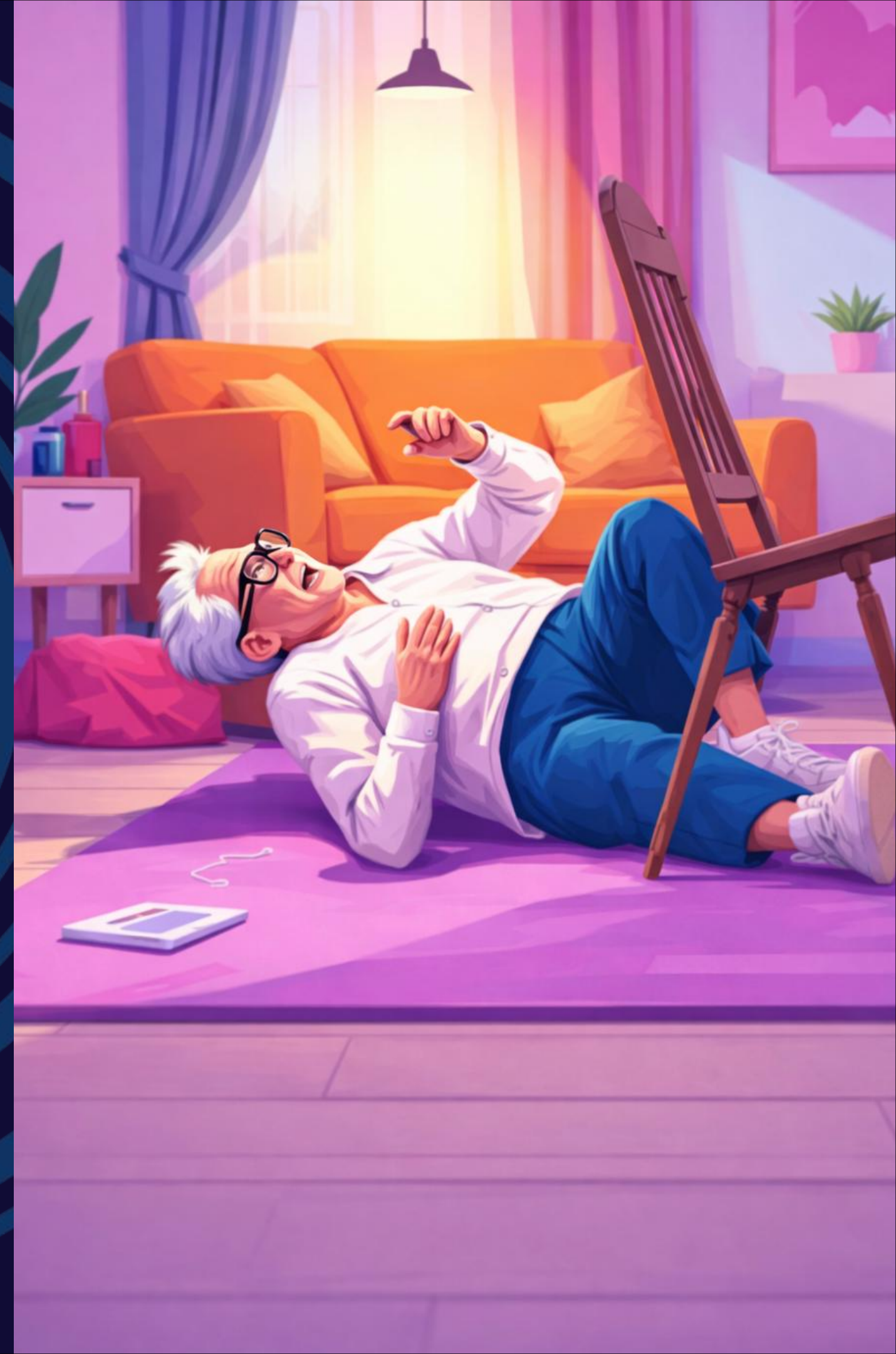
Elderly individuals face daily fall hazards due to balance issues, medication effects, and declining muscle strength — often when no one is nearby.

🕒 Delayed Emergency Response

Without automatic detection, victims may remain unattended for hours, dramatically worsening outcomes and increasing mortality risk.

🚫 No Automated System

Existing solutions are largely manual, expensive, or require constant human supervision — making them impractical for independent living.



Project Objectives

Detect Falls Automatically

Use accelerometer and gyroscope data to identify sudden falls with high accuracy in real time.

Instant On-Watch Alert

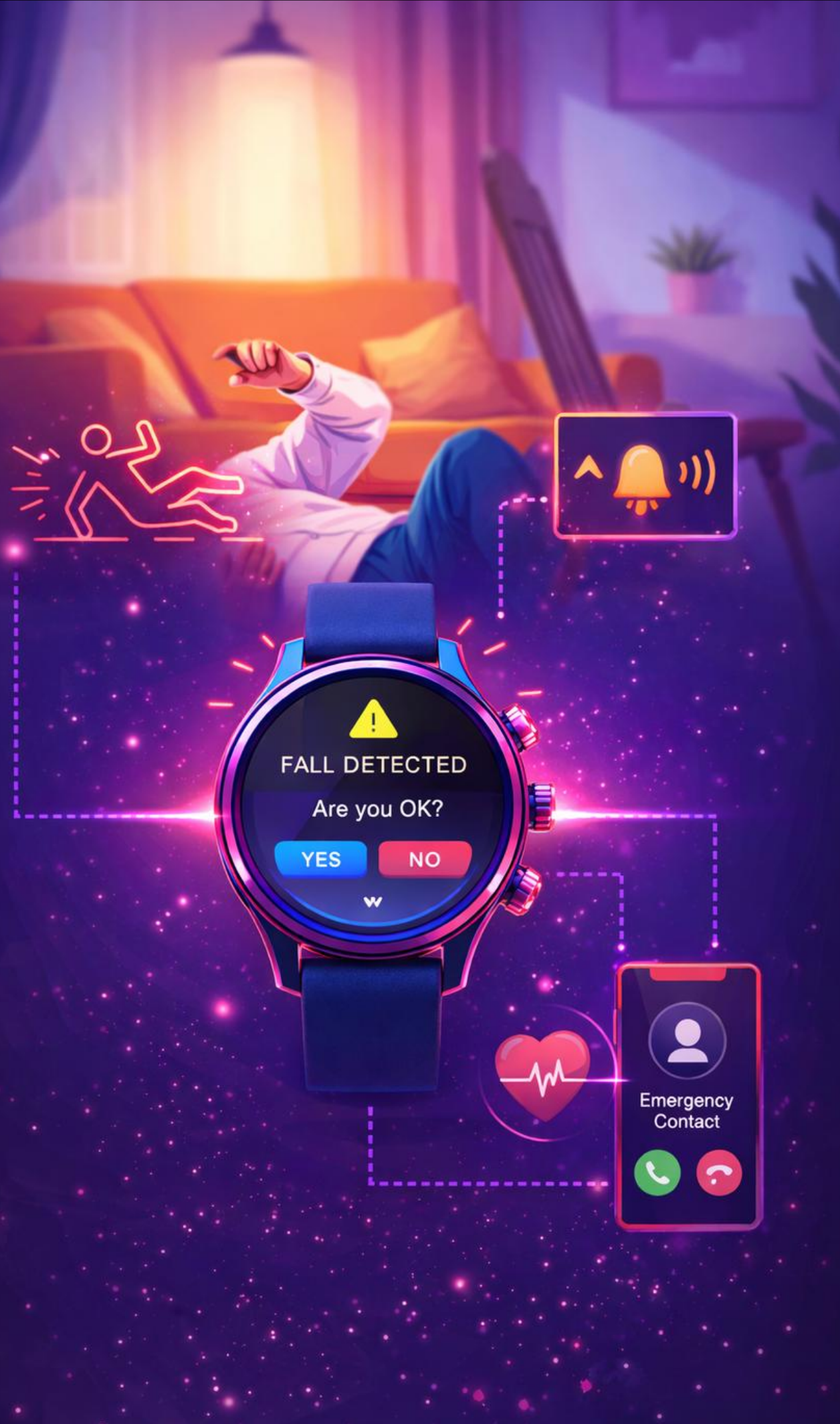
Trigger an immediate popup and buzzer on the smartwatch to confirm user safety within seconds.

Send Emergency Alert

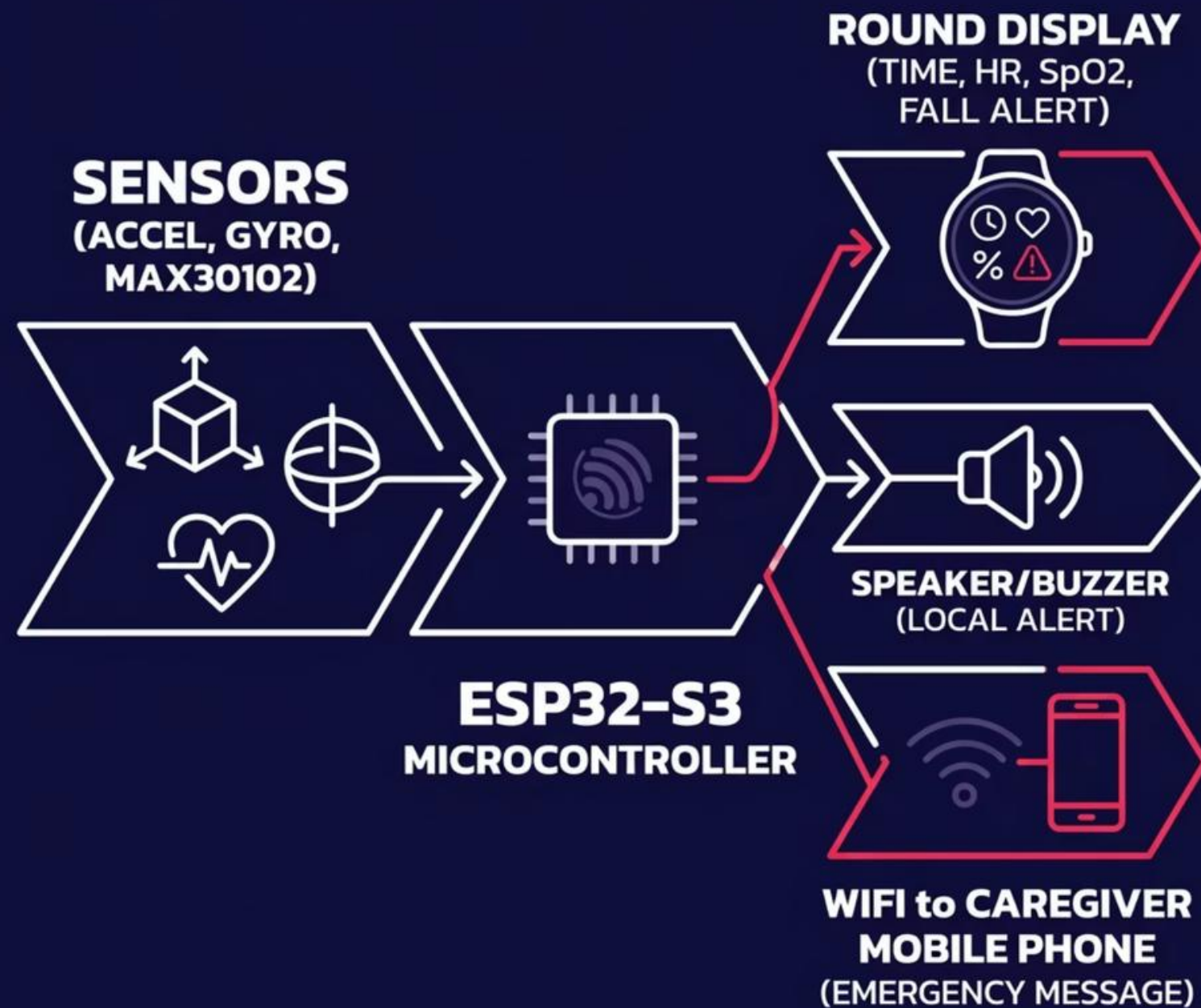
Automatically notify a designated caregiver via mobile message if the user fails to respond.

Monitor Vitals

Continuously measure heart rate and blood oxygen (SpO2) to assess post-fall health status.



System Architecture



How the System Connects

All sensing, processing, and communication happens on the wrist. The ESP32-S3 acts as the central brain, aggregating sensor data, managing the display, triggering local alerts, and dispatching emergency notifications — all wirelessly and in real time.

 Wireless
Wi-Fi via ESP32

 Real-Time
Instant processing

 Wearable
Compact form factor

Hardware & Software Stack

Hardware Components

1

ESP32-S3 Waveshare Round Display

Main processing unit with integrated circular display and Wi-Fi/Bluetooth

2

Accelerometer & Gyroscope (In-Build IMU)

Detects rapid changes in orientation and acceleration for fall identification

3

MAX30102 — Heart Rate & SpO2

Optical sensor measuring pulse and blood oxygen saturation post-fall

Software Tools

Arduino IDE

Embedded C programming environment for ESP32 firmware development

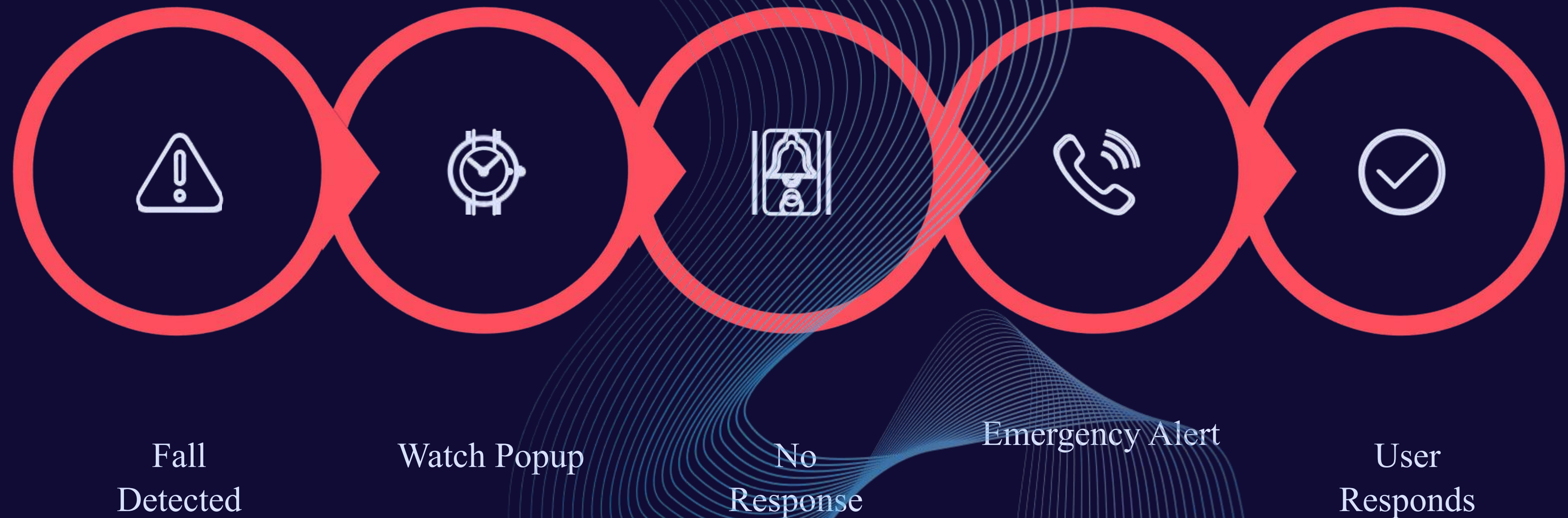
LVGL / Custom GUI

Lightweight graphics library powering the smartwatch circular display interface

Wi-Fi & HTTP Libraries

ESP32 native libraries for wireless connectivity and emergency message dispatch

Working Principle — Fall Detection Logic



The decision tree is fully autonomous: from fall detection through vital sign assessment to emergency dispatch — all within 15 seconds, requiring zero manual intervention from the user or caregiver.

Smartwatch GUI Design

Time Display

Clean clock face with date —
always-on ambient mode for
easy glancing

Heart Rate &

SpO2

Live vitals shown on the
home screen with color-coded
health status indicators

Fall Alert

Popup

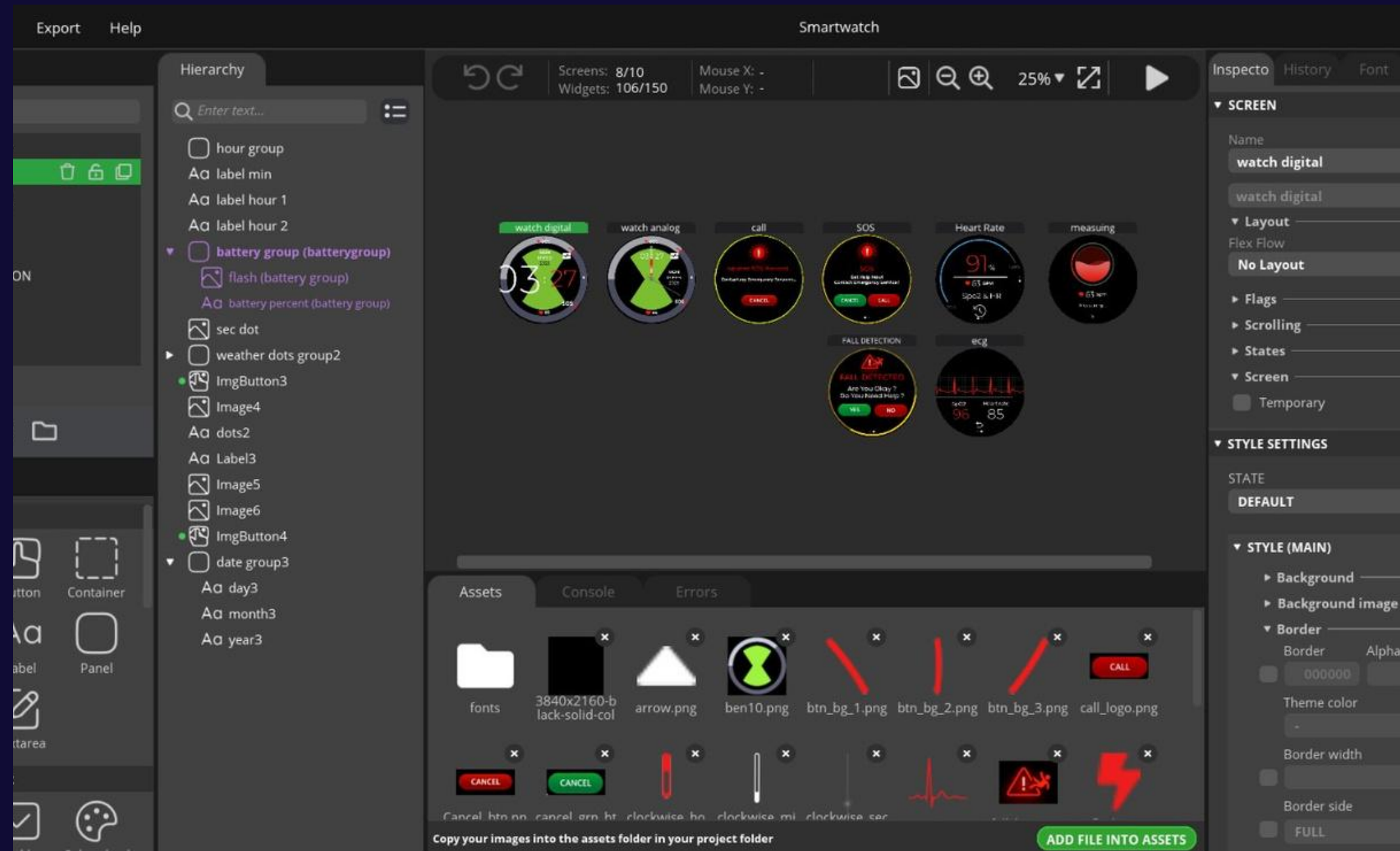
"Are you OK?" full-screen overlay with a single-tap confirm button
to cancel the alert



software

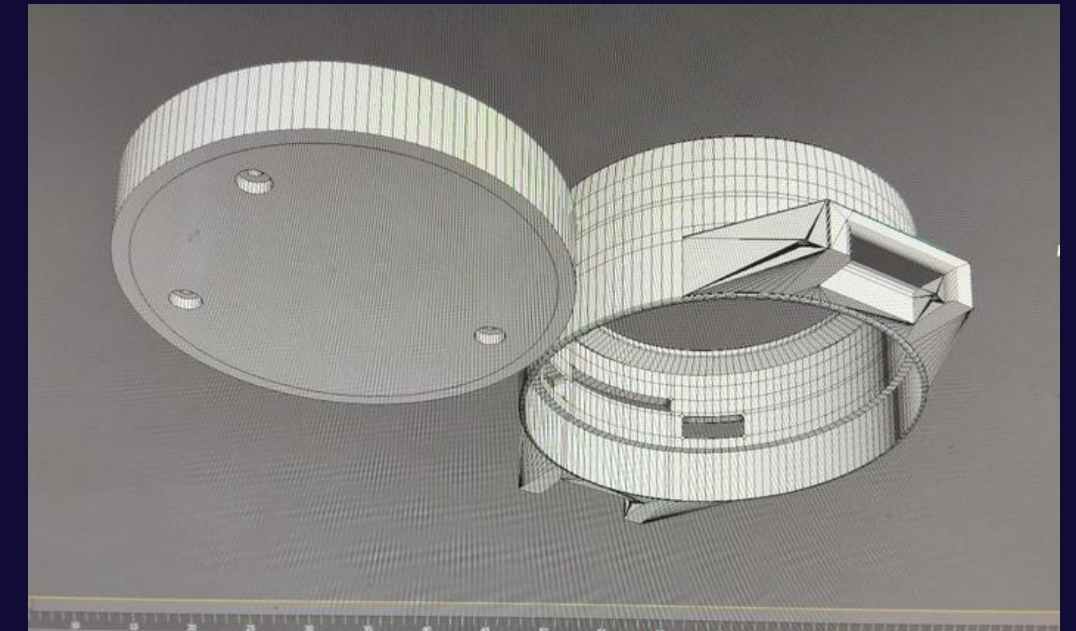
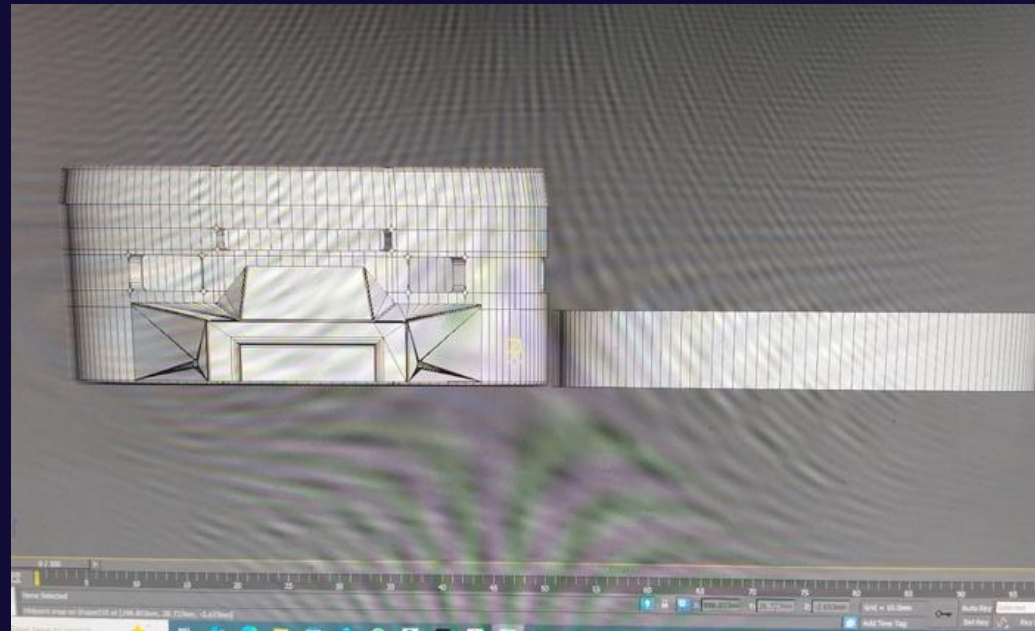
square mile studio

The user interface is developed using SquareLine Studio and deployed with LVGL to visualize real-time output on the display.



Implementation on software :

3D modeling using 3d max software



The watch model is developed in Autodesk 3ds Max

Implementation on Hardware :



LVGL / Custom GUI

The system uses LVGL to design a custom GUI, where real-time output is visualized on the display.

Features, Advantages & Applications

✓ Key Features & Advantages

- Real-Time Fall Detection
 - Detects falls using IMU sensors (Accelerometer + Gyroscope)
 - Provides sub-second response time
 - Differentiates between normal movement and actual falls
 - Ensures quick emergency detection
- Automated Emergency Alerts
 - Automatically sends SMS/Call via GSM module
 - No user intervention required
 - Alerts family members and caregivers instantly
- Reduces delay in emergency response
- Cost-Effective & Wearable
 - Built using low-cost components
 - Affordable compared to existing medical devices
 - Compact, lightweight, and easy to wear
- Suitable for home and hospital use
- Continuous Health Monitoring
 - Monitors heart rate and SpO₂ levels
 - Provides real-time health tracking
 - Helps in early detection of health issues
 - Suitable for long-term patient

🌐 Real-World Applications

- Elderly Home Care
 - Enables independent living for senior citizens
 - Provides continuous monitoring
 - Sends alerts in case of falls or emergencies
- Hospitals & Medical Wards
 - Useful for post-operative and high-risk patients
 - Prevents unattended falls
 - Assists healthcare staff with monitoring
- Rehabilitation Centers
 - Tracks patient recovery and movement
 - Helps physiotherapists monitor progress
 - Improves rehabilitation outcomes

Future Scope & Conclusion

Future Enhancements



GPS Tracking

Real-time location sharing for outdoor fall scenarios



AI-Based Fall Prediction

Machine learning models to anticipate falls before they occur



Mobile App Integration

Cloud Storage & Analytics



Caregiver dashboard for live vitals, history, and alerts

Long-term health trend analysis and incident reporting

Conclusion

This project presents an IoT-based wearable system for real-time fall detection and health monitoring, aimed at enhancing elderly safety. By integrating sensors and GSM communication, it provides instant alerts without user intervention. The system is compact, cost-effective, and suitable for both home and hospital use, offering a reliable and scalable solution with the potential to save lives.



Reference papers

[1] Gupta, S. N., Hussain, A., Kumar, A., & Pramanik, I. (2024). IoT based Smart Fall Detection Device for Elderly People. 2024 International Conference on Electrical Electronics and Computing Technologies (ICEECT), 1-5.
<https://doi.org/10.1109/iceect61758.2024.10739038>

[2] Krishna, A., Yadav, B., Prakash, D., Dwivedi, R. K., & IEEE. (2024). Designing A Fall Detection System for Elderly People in Smart Healthcare IoT Using Multi Sensor Data Fusion. 2024 International Conference on Artificial Intelligence and Quantum Computation-Based Sensor Application (ICAIQSA).
<https://doi.org/10.1109/ICAIQSA64000.2024.10882340>

[3] J. Akilandeswari and A. Velusamy, "Recent Advances in Fall Detection and Prevention Using Machine Learning and IoT: A Comprehensive Review," in Proc. Int. Conf. on Emerging Technologies in Engineering Applications

